THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $0.\overline{18} \times 11$ equals: 1

(a) 3 **(b)** 2 **(c)** $\frac{2}{3}$ **(d)** 1

2. $8x^3 + 27$ factorises as: 1

(a) $(2x+3)^3$ **(b)** $(2x+3)(4x^2+6x+9)$ **(c)** $(2x+3)(4x^2-6x+9)$ **(d)** $(2x-3)(4x^2+6x+9)$

3. The area of the triangle with vertices $(0,0)$, $(18,0)$ and $(0,8)$ is: 1

(a) 74 **(b)** 71 **(c)** 72 **(d)** 73

4. The area (in square units) of the triangle formed by the line $y = 4x + 12$ and the two axes is: 1

(a) 18 **(b)** 17 **(c)** 19 **(d)** 20

5. The value of $0.\overline{4} + 0.\overline{8}$ is: 1

(a) $\frac{4}{3}$ **(b)** $\frac{5}{3}$ **(c)** 1 **(d)** $\frac{4}{5}$

6. If $x + y = 7$ and $xy = 10$, then $x^3 + y^3 =$ 1

(a) 133 **(b)** 135 **(c)** 134 **(d)** 132

7. The distance of the circumcentre of the right triangle with vertices $(0,0)$, $(10,0)$ and $(0,24)$ from the origin is: 1

(a) 15 **(b)** 14 **(c)** 12 **(d)** 13

8. The x-coordinate of the point where $y = 7x - 2$ and $y = 3x + 18$ meet is: 1

(a) 4 **(b)** 7 **(c)** 6 **(d)** 5

9. $0.\overline{26} + 0.\overline{62}$ equals: 1

(a) 1 **(b)** $\frac{4}{5}$ **(c)** $\frac{7}{9}$ **(d)** $\frac{8}{9}$

10. If $x - y = 3$ and $xy = 10$, then $x^3 - y^3 =$ 1

(a) 118 **(b)** 117 **(c)** 119 **(d)** 116

11. A circle has diameter with endpoints $(-3,4)$ and $(21,74)$. Its radius is: 1

(a) 38 **(b)** 39 **(c)** 36 **(d)** 37

12. If $p(x) = 3x - 1$, then $p(p(2)) =$ 1

(a) 16 **(b)** 15 **(c)** 14 **(d)** 13

13. **Assertion (A):** The number $\sqrt{2}$ is irrational. 1
Reason (R): The decimal expansion of an irrational number is non-terminating and non-repeating.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** $x^2 - 9 = (x-3)(x+3)$. 1
Reason (R): $(a+b)^2 = a^2 + 2ab + b^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The point $(4, -2)$ lies in Quadrant IV. 1
Reason (R): In Quadrant IV both the abscissa and the ordinate are negative.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If $x = 6 + \sqrt{35}$, find $\frac{1}{x}$ and $x + \frac{1}{x}$. 2

17. Factorise $8x^3 + 1$. 2

18. The vertices of a right triangle are $O(0,0)$, $A(10,0)$ and $B(0,24)$. Find the circumcentre and the circumradius. 2

19. Plan A costs ₹150 plus ₹25 per unit; Plan B costs ₹390 plus ₹5 per unit. For how many units do both cost the same? 2

20. Express $0.\overline{29} + 0.\overline{92}$ as a single fraction in lowest terms. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Express $5.3\overline{2}$ as a fraction in lowest terms and verify by long division. 3

22. If $x + \frac{1}{x} = 8$, find $x^3 + \frac{1}{x^3}$. 3

23. Three vertices of a rectangle $ABCD$ are $A(1,2)$, $B(13,18)$ and $D(-11,11)$. Find C and the diagonal AC. 3

24. A linear polynomial has $p(1) = -10$ and $p(3) = 0$. Find $p(x)$, its zero and $p(4)$. 3

OR

If $x = 3 + \sqrt{8}$, find $x^2 + \frac{1}{x^2}$. 3

25. Factorise $x^2 - 9x + 14$ and use it to find the value when $x = 10$. 3

OR

Find the length of the median AD of the triangle with $A(0, 0)$, $B(48, 0)$, $C(0, 64)$, where D is the midpoint of BC. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Mrs Gupta writes the fractions $\frac{7}{16}$, $\frac{5}{12}$ and $\frac{11}{25}$ on the board and asks the class to find their decimal expansions. 5

(i) Write $\frac{7}{16}$ as a decimal.

(ii) Write $\frac{11}{25}$ as a decimal.

(iii) Express the denominator 12 in prime factors.

(iv) Which of the three fractions have terminating decimal expansions? Give the reason.

27. Find the decimal expansion of $\frac{5}{13}$ by long division, state the repeating block and its length, and hence write $\frac{5}{13}$ as a repeating decimal. 5

OR

If $x - y = 2$ and $xy = 24$, find $x^2 + y^2$, $x^3 - y^3$ and $(x + y)^2$. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND